

“When I thought about life, I found it tragic.
When I felt it, I found it beautiful.”

— Charlie Chaplin

A Gentle Way of Thinking for Noise-Sensitive Individuals

L. Xie

Independent author

yanxi.xie.contact@gmail.com

February 17, 2026

Abstract

This paper offers a first-person, exploratory reflection on thinking, learning, and expression from the perspective of noise-sensitive individuals. Rather than proposing a general theory or methodological prescription, it examines how experiential noise, coherence, and dimensionality may interact during the process of reasoning.

The paper introduces the notion of a high-dimensional but finite experiential space, in which thinking, reasoning, and language can be understood as gradual compression processes guided by coherence. Within this view, coherence is treated not as a static property, but as something that may remain engaged and be repeatedly re-aligned as thinking unfolds. Language, including mathematics, is discussed as a representational structure that may support this process, with mathematics described as a particularly reliable and internally consistent structure that allows coherence to be re-aligned during reasoning.

Several tentative reflections are explored, including the role of first-person experience, intuition, uncertainty, learning pace, cross-disciplinary perspectives, and the limits of early formalization. These reflections are not presented as conclusions, but as possible ways of understanding how thinking may remain sustainable under heightened experiential sensitivity.

The aim of this paper is not to provide ready-to-use answers, but to offer support and orientation. If the ideas presented here help noise-sensitive readers maintain confidence while continuing to think carefully, they may be taken as such. No agreement or adoption is required.

1 Experiential Noise and the Noise Space $E(t)$

In my experience, thinking is not a process that occurs purely at the level of abstract reasoning. It is always accompanied by a background experiential state that evolves over time: tension, openness, pressure, relaxation, urgency, or spaciousness.

I distinguish two related but different notions.

First, I refer to the structural context in which experience occurs as the *noise space*, denoted by

$$E(t).$$

The noise space does not correspond to a specific emotion. Rather, it describes the experiential terrain that determines whether disturbances tend to accumulate or dissipate.

Second, I refer to the directly felt discomfort or interference within this space as *experiential noise*, denoted by

$$f_{\text{noise}}(t).$$

When $E(t)$ is unstable, even minor stimuli may cause $f_{\text{noise}}(t)$ to rise rapidly. When $E(t)$ is relatively stable, experiential noise becomes easier to carry.

For me, an important and often overlooked fact is that experiential noise is not a side effect of thinking, but a foundational condition for whether thinking can be sustained.

2 Coherence Detection as a Multidimensional Scan of $E(t)$

Over time, I began to notice that changes in experiential noise are not determined solely by external conditions. They often arise from how I internally relate to and evaluate my own thinking process.

I refer to this ongoing internal evaluation as *coherence detection*. It is not a judgment of correctness, but rather an implicit and continuous question: whether the current way of thinking remains compatible with the experiential structure I am able to carry at that moment.

Formally, I denote this process as

$$C(E(t)).$$

When coherence is maintained across multiple dimensions, experiential noise tends to decrease over time:

$$\frac{df_{\text{noise}}(t)}{dt} < 0.$$

A reflective note on the role of reason. In further reflection, I have noticed a subtle phenomenon that seems particularly relevant for noise-sensitive individuals. During the thinking process, reason does not appear to function well as an independent or leading force. Instead, it seems to need to *follow coherence closely* and to remain continuously engaged in the coherence-detection process itself.

When rational reasoning stays aligned with coherence and moves together with it, thinking often remains relatively stable, and experiential noise tends to stay within a carryable range. However, when reason attempts to advance on its own—detached from coherence and prior to it—the thinking process may begin to distort. Such distortion does not necessarily manifest immediately as a logical error; it may first appear at the experiential level, for example as discomfort, tension, or a loss of directional sense.

This observation does not suggest that reason is problematic, nor that coherence should replace rational analysis. Rather, it may indicate that, for noise-sensitive individuals, the role of reason within thinking is itself position-dependent: reason appears to need to participate continuously in coherence detection, rather than operating ahead of or independently from it.

At present, this remains an experiential and tentative observation. Whether it admits a more general explanation, or applies across different modes of thinking, is left open for further exploration by others.

3 High-Dimensional Experiential Space, Coherence-Driven Compression, and Noise Dynamics

To better understand the thinking process of noise-sensitive individuals, I introduce a structural assumption: experience initially unfolds in a *high-dimensional but finite* space.

Here, high dimensionality should not be interpreted as infinite dimensionality. Human perception is inherently limited, and it therefore seems more accurate to regard the experiential space $\mathcal{S}_{\text{exp}}$ as having a relatively high, but finite and unknown, number of dimensions. The notion of high dimensionality is used only to indicate that experiential structure typically involves many parallel and not-yet-differentiated factors operating simultaneously.

From this perspective, thinking, reasoning, creation, and verbalization are not the starting point of experience, but can be understood as a process that gradually compresses high-dimensional experience into lower-dimensional, carryable forms:

$$\mathcal{C}_t : \mathcal{S}_{\text{exp}} \longrightarrow \mathcal{S}_{\text{thought}}, \quad \dim(\mathcal{S}_{\text{thought}}) \ll \dim(\mathcal{S}_{\text{exp}})$$

For me, this compression process does not occur arbitrarily. Rather, it appears to be guided by coherence detection operating in parallel across multiple dimensions:

$$\mathcal{C}_t \equiv C(E(t))$$

When this coherence-driven compression proceeds at an appropriate rate and scope, experiential noise often exhibits a sustained downward trend over time:

$$\frac{df_{\text{noise}}(t)}{dt} < 0$$

In my own reasoning practice, I tend to experience mathematics as a structure that, during the process of reasoning, continually allows coherence to be re-aligned. This description reflects a personal way of engaging with mathematics, rather than a general claim about its nature.

From this perspective, coherence is not treated as something fixed in advance, but as something that can be repeatedly located and adjusted as reasoning unfolds. The high degree of internal consistency that characterizes mathematics means that, when derivations are carried out correctly, internal error can in principle be zero. For this reason, mathematics remains one of the most reliable representational tools available, particularly for maintaining precision, stability, and shared reference across different contexts.

An intuitive note on coherence. Here, coherence is not meant to imply uniformity of method or viewpoint. It may be understood instead through a simple metaphor: different individuals may begin from different locations and take different paths up a mountain, using different tools, routes, or strategies. As long as they are oriented toward the same summit, they may eventually meet there, despite having taken distinct trajectories.

In this sense, coherence does not require identical reasoning processes, but a shared directional alignment. Mathematics, viewed as a structure that, during the process of reasoning, continually allows coherence to be re-aligned, may be seen as one way of expressing this alignment explicitly, even when the paths leading to it differ.

I emphasize that this section reflects a personal way of organizing experience, language, and formal reasoning, rather than a claim or prescription. It does not require agreement or acceptance. If these distinctions prove useful, they may be taken as such; if not, they may be set aside.

4 Conditions That Disrupt the Process

Within the framework described above, I have also observed several conditions that tend to systematically disrupt coherence-driven compression for noise-sensitive individuals. These conditions are not inherently incorrect, but they interact poorly with this particular experiential structure.

A common failure mode occurs when thinking, reasoning, or creative structuring precedes experience. When concrete methods, data, metrics, or proofs are introduced too early, the high-dimensional experiential space is prematurely projected onto low-dimensional, highly determinate representations. As a result, the solution space may become frozen before it has been adequately explored.

Importantly, avoiding premature freezing does not imply abandoning structure. For noise-sensitive individuals, a safer early-stage strategy is often to attend to *constraints before solutions*. Identifying what may not work, what cannot be violated, or what remains incompatible with the current experiential structure does not collapse the solution space. Instead, it gently bounds that space in a way that remains carryable.

At the level of language, this approach is often accompanied by deliberate use of uncertainty markers such as “may,” “might,” “it seems,” or “under current conditions.” These expressions do not weaken reasoning. Rather, they preserve room for solutions that have not yet emerged, preventing premature commitment to a single low-dimensional path.

From this perspective, well-known approaches such as inverse thinking or first-principles reasoning may derive their gentleness from a shared feature: they constrain what does *not* hold before asserting what must hold. This observation is specific to noise-sensitive individuals and is not intended as a general prescription.

5 Conditions That Support Gentle Compression

5.1 A Carryable External Environment

For noise-sensitive individuals, environments that support gentle compression tend to share several characteristics: the absence of urgent task pressure, the lack of rigid deadlines, and the absence of demands for immediate convergence or proof.

In such environments, experience is allowed to remain high-dimensional for a sufficient period of time. Thinking, reasoning, and creation can then proceed under coherence constraints rather than external compression forces, allowing gradual, stable reduction of experiential noise.

5.2 Independent Thinking and Invisible, Constructive Collaboration

Noise-sensitive individuals often benefit from extended periods of independent thinking. Forced or tightly synchronized collaboration can introduce external low-dimensional constraints before experiential integration has occurred.

More supportive forms of collaboration tend to be invisible or constructive: non-immediate feedback, asynchronous exchange, and interactions that illuminate structure rather than impose direction. Historical mathematical correspondence provides a gentle example of such collaboration, offering external perspective without violently freezing the solution space.

5.3 Low-Noise Universal Tools and Strong Noise Gradients

At appropriate stages of compression, certain universal tools can dramatically reduce experiential load. These tools compress highly complex, dynamic, and effectively high-dimensional structures into representations that are low-dimensional, static, and structurally complete.

Abstractly, such tools can be expressed as

$$\mathcal{T} : \mathcal{S}_{\text{exp}} \longrightarrow \mathcal{R}, \quad \dim(\mathcal{R}) \ll \dim(\mathcal{S}_{\text{exp}}).$$

When these tools align well with the current noise space $E(t)$, they often induce a strong experiential gradient:

$$\frac{df_{\text{noise}}(t)}{dt} \ll 0.$$

For noise-sensitive individuals, tools such as Newton's laws, Maxwell's equations, Bode plots, and the Fourier transform function not only as analytical instruments, but also as highly efficient noise-reducing structures. Once established, they can sustain a long-term downward trend in experiential noise without repeated high-dimensional effort.

5.4 High-Dimensional Expression, Everyday Language, and Flexible Compression

After considering the role of low-noise universal tools, I would like to add one further observation related to expression and language.

In the thinking space described above, higher-dimensional regions often appear to be associated with poetic, intuitive, or everyday forms of expression. By contrast, highly abstract and precise terminology tends to appear in lower-dimensional representations. At the same time, higher-dimensional spaces typically contain lower-dimensional ones, rather than excluding them.

This suggests that complex knowledge may often be expressible using simple, everyday, or even poetic language. When a complex idea can be explained accurately through concise and familiar expressions, such expression does not merely improve communicability; it may also preserve a certain degree of high-dimensional flexibility.

From this perspective, a “good” expression may be described as one that compresses a complex experiential structure while retaining access to its higher-dimensional degrees of freedom. Schematically, this can be written as

$$\mathcal{E}_{\text{good}} : \mathcal{S}_{\text{exp}} \longrightarrow \mathcal{L}_{\text{simple}}, \quad \text{with residual flexibility preserved in } \mathcal{S}_{\text{exp}}.$$

Here, $\mathcal{L}_{\text{simple}}$ denotes a space of simple, everyday, or intuitive language, while the mapping does not exhaust or rigidly fix the original experiential structure. Such expressions may therefore remain revisitable, adaptable, and expandable.

This observation may also help explain why learning approaches that emphasize explanation in one’s own simple words—such as the so-called Feynman learning technique—often feel effective. By attempting to explain a concept using ordinary language, one is not merely checking recall, but testing whether the compression preserves coherence without prematurely collapsing the underlying structure.

As with the rest of this section, this observation is offered tentatively. It is not a claim about optimal pedagogy in general, but a possible alignment between language, dimensionality, and noise-sensitive thinking.

6 Extended Reflections and a Tentative Conclusion

This section is written primarily for noise-sensitive individuals. Its purpose is not to establish new claims or to persuade others, but to offer a possible source of confidence in continuing to think. If any of these reflections seem useful, they may be taken as such; if not, they need not be adopted.

I begin with a speculative observation about the structure of the thinking space itself. In the framework used throughout this paper, both the extremely high-dimensional limit and the zero-dimensional limit may correspond to what is, in different ways, unsayable.

The high-dimensional end of the thinking space appears to involve rich, parallel, and pre-

linguistic structure. At the same time, the zero-dimensional limit may correspond to an extreme form of precision in which no internal degrees of freedom remain. In this sense, both ends may resist direct linguistic expression.

This speculation loosely echoes Wittgenstein’s suggestion that the limits of language are the limits of one’s world. If so, a zero-dimensional representation might be maximally precise, yet precisely because of that precision, it may no longer admit meaningful language at all. From this perspective, whenever we use language—possibly even mathematical language—we may already be carrying a certain degree of indeterminacy.

I emphasize that this is a personal and tentative conjecture, not a conclusion. Its role here is not explanatory authority, but to leave space for further thought by others.

6.1 A tentative conjecture on first-person thinking and coherence

If the above speculation holds even partially, then first-person thinking by noise-sensitive individuals may not be low-quality thinking simply because it remains close to experience. Its apparent indeterminacy may reflect the structure of the thinking space itself, rather than a lack of rigor.

From this point of view, what may matter more than the elimination of uncertainty is whether coherence continues to guide and participate in the thinking process. As long as coherence remains engaged and aligned, first-person reflection may remain stable, meaningful, and capable of genuine insight, even if it does not immediately crystallize into low-dimensional form.

This conjecture does not claim general validity. It is offered as a possible re-framing: for noise-sensitive individuals, difficulty in early formalization may not indicate failure, but a different relationship between language, dimensionality, and experience.

The intention of this section is simply to leave room. Room to think without rushing, room to remain uncertain without self-dismissal, and room to continue engaging with complex problems without assuming that one’s starting point is illegitimate.

A further reflection may be added here.

First-person thinking may not, in itself, imply low-quality thinking. What seems more relevant is whether coherence continues to guide and participate in the thinking process over time. When coherence remains engaged, first-person reflection may stay stable and internally consistent, even if it initially lacks formal structure or external validation.

At the same time, this does not preclude other forms of verification. After a period of thinking or exploration, it may be possible—and sometimes helpful—to look across dimensions for more rigid forms of support, such as empirical data, formal analysis, or experimental results. Such cross-dimensional checks need not replace first-person reasoning; they may instead serve as a secondary alignment step, providing additional stability and reinforcing coherence.

From this perspective, coherence-guided first-person thinking and later cross-dimensional validation do not appear to be in conflict. They may function as different phases of the same process: one oriented toward exploration and internal alignment, the other toward consolidation and shared

reference. This sequence may allow noise-sensitive individuals to maintain both experiential stability and a sense of epistemic grounding.

6.2 A tentative note on expertise, coherence, and entry into a field

Another possible reflection concerns the role of professional knowledge in relation to thinking itself.

Professional or domain-specific knowledge may not always be a strict prerequisite for beginning to think meaningfully about a field. What seems more decisive, especially for noise-sensitive individuals, is whether coherence continues to guide and participate throughout the thinking process. When coherence remains engaged, initial exploration may remain stable and oriented, even in the absence of extensive formal training.

This is not to suggest that professional knowledge is unimportant. On the contrary, foundational knowledge in a domain often provides crucial structure, helps avoid unnecessary confusion, and can significantly reduce experiential noise. Such foundations may support coherence rather than replace it.

Moreover, deeper professional expertise may be necessary for meaningful integration between a field and everyday life. To offer reliable insight, practical contribution, or qualified service to others, one often needs to be sufficiently grounded in the domain itself. In this sense, professional depth may be a condition for responsible application rather than for the initial legitimacy of thinking.

From this perspective, coherence-guided thinking and professional knowledge do not appear to be in opposition. They may instead play complementary roles: coherence providing internal alignment and sustainability, and expertise providing structure, precision, and the ability to connect thought with shared practice.

6.3 A tentative note on first-person language and effective communication

One further reflection concerns language itself, particularly the role of first-person, everyday forms of expression.

First-person and life-oriented language may, in some cases, function as a more effective medium for communication. Such language seems to align more easily at the experiential level, allowing others to resonate not only with the content of what is said, but also with how it is felt. In this sense, communication may succeed not through maximal precision alone, but through a shared alignment in the experiential layer.

At the same time, this approach may involve trade-offs. In highly technical or specialized contexts, first-person or everyday language may lose a degree of precision, and certain distinctions may become harder to maintain. This does not imply that such language is inadequate, but rather that it may not always be sufficient on its own.

From this perspective, a possible goal is not to privilege either everyday expression or technical formulation exclusively, but to search for an intermediate position. Such a position might allow communication to remain efficient and experientially aligned, while still preserving enough precision to avoid misunderstanding or loss of structure.

This balance may not be fixed. It may depend on context, audience, and purpose. For noise-sensitive individuals, being aware of this spectrum may itself provide flexibility: the ability to move between registers, rather than being constrained to a single mode of expression.

6.4 A tentative reflection on intuition, logic, and guidance

One further reflection concerns intuition and its relationship to logic.

Intuition does not necessarily imply an absence of logic. For noise-sensitive individuals, intuition may instead function as a form of coherence detection that operates across multiple dimensions in parallel, possibly spanning different temporal or contextual scales. Rather than proceeding step by step, intuition may resemble a kind of “skipped-step” alignment, where coherence is sensed without explicit intermediate articulation.

From this perspective, intuition may sometimes serve as a useful form of directional guidance. It may indicate regions of the thinking space where coherence appears stronger, even if the underlying structure has not yet been fully articulated in logical or formal terms.

This observation does not suggest that rigorous reasoning is unnecessary. On the contrary, explicit reasoning and formal analysis may provide a stronger sense of solidity, clarity, and shared reference. Such rigor can help stabilize insights, test intuitions, and support communication across different contexts.

The possible distinction, then, may not be between intuition and logic, but between different phases or roles within the thinking process. Intuition may offer orientation or initial direction, while rigorous reasoning may contribute consolidation, verification, and durability. For noise-sensitive individuals, recognizing both roles may allow thinking to remain both flexible and grounded.

6.5 A tentative note on learning pace and individual methods

A further reflection concerns learning speed and individual learning methods.

Learning slowly, or learning in a highly personal way, does not necessarily indicate a lack of ability. In some cases, it may reflect the need for a longer period of high-dimensional scanning, during which coherence is repeatedly checked before new knowledge can be integrated into an existing structure.

For noise-sensitive individuals, new concepts may need to be examined from multiple experiential angles and across different contexts before they can be accepted as internally consistent. This process may take more time, but it may also allow the resulting understanding to be more stable and less prone to internal conflict.

From this perspective, a slower pace of learning may not represent inefficiency, but a different allocation of effort: more time spent on coherence confirmation in higher-dimensional space, followed by a more durable compression into lower-dimensional form. This possibility does not imply superiority, nor does it generalize to all learners, but it may offer an alternative interpretation to narratives that equate speed with intelligence.

As elsewhere in this section, this reflection is offered tentatively. If it resonates with personal experience, it may be taken as such; if not, it requires no adoption.

6.6 A tentative note on cross-disciplinary learning and structural insight

A further personal observation concerns cross-disciplinary learning.

For noise-sensitive individuals, engaging with multiple disciplines may make it easier to notice the *structure* of knowledge itself, rather than only its surface content. When similar patterns, constraints, or relations appear across different domains, coherence may become more salient, even if each individual perspective remains partially uncertain.

This situation may be loosely compared to practices in scientific measurement. In many measurement contexts, a quantity is not determined from a single dimension or observation, but through cross-validation from multiple independent directions. Each measurement may carry a degree of uncertainty, yet the intersection of several such constraints can locate a point more reliably than any single line alone.

In this analogy, cross-disciplinary perspectives function like multiple lines, each with limited precision, but together providing stronger structural localization. From this view, learning across disciplines may help noise-sensitive individuals confirm coherence from several directions, rather than relying on a single pathway of understanding.

I emphasize that this is only a personal conjecture. It is offered as a possible way of thinking about learning and structure, not as a general claim or prescription. If it proves useful, it may be taken as such; if not, it can be left aside for others to consider.

6.7 A tentative note on the purpose of thinking and unavoidable participation

One further reflection concerns the purpose of thinking, especially at its early stages.

The aim of thinking may not always be to provide a highly reliable or immediately usable answer. In particular, high-dimensional, intuitive, and experience-laden thinking often carries substantial uncertainty. Imagination itself seems able to operate most freely in such high-dimensional regions, and its range may narrow prematurely when fear of error becomes dominant.

For writers and thinkers, excessive concern about being wrong may lead to paralysis at the very beginning of the thinking process. When early exploration is constrained too tightly, movement within the thinking space may become difficult, and potential directions may remain unexplored.

Another personal observation is that, during the process of confirming an answer, one's own thinking appears to be unavoidable. No external authority, person, or tool can fully substitute for this internal confirmation. Even when external references are available, the act of alignment and acceptance still seems to require personal participation.

From this perspective, the purpose of thinking may sometimes be not to deliver a ready-made, correct, and convenient answer, but to invite others to willingly engage in this unavoidable process of thinking. In such cases, the value may lie less in the final form of the answer and more in the shared act of exploration itself.

Conclusion

This paper has been written primarily for noise-sensitive individuals. It does not aim to establish a theory, define standards of correctness, or provide a set of reliable, ready-to-use answers. Instead, it has attempted to describe a possible way of approaching thinking, learning, reasoning, and expression that may remain sustainable under conditions of heightened experiential sensitivity.

Throughout the text, first-person experience has not been treated as a guarantee of correctness, nor as a substitute for rigor, expertise, or verification. Rather, it has been approached as a possible point of entry: a place where coherence may be sensed, where it may continue to guide and participate in thinking, and where uncertainty can be tolerated without immediate collapse. For noise-sensitive individuals, the ongoing engagement of coherence may matter more than the speed at which conclusions are finalized or formalized.

Several tentative reflections have been explored. High-dimensional experience may require time before it can be compressed into stable forms. Language, including mathematics, may function not only as a representational tool, but as a structure that allows coherence to be repeatedly realigned during reasoning. Intuition may provide directional guidance, while explicit reasoning and cross-dimensional checks may offer consolidation and stability. Professional knowledge may deepen responsibility and applicability, even if it is not always a prerequisite for meaningful initial exploration. Learning slowly, or through personal methods, may reflect extended coherence confirmation rather than lack of ability. Cross-disciplinary perspectives may help reveal structure by allowing coherence to be confirmed from multiple directions. None of these observations are presented as general claims; they are offered as possible ways of understanding experience and thought.

From this perspective, the purpose of thinking—especially in its early stages—may not be to deliver highly reliable answers to others. High-dimensional, intuitive, and imaginative thinking often carries unavoidable uncertainty, and fear of error may restrict exploration prematurely. Moreover, the act of confirming understanding appears to require personal participation: no external authority, person, or tool can fully replace one’s own process of alignment.

In some cases, then, the value of thinking may lie less in producing immediately correct or usable results, and more in enabling voluntary engagement with an unavoidable process. If this text offers noise-sensitive readers a measure of confidence to continue thinking, to tolerate uncertainty, and to participate without self-dismissal, then it has served its purpose. If not, it requires no agreement or adoption.

This paper does not claim necessity. It simply leaves a possible map for those who may recognize the terrain.

Acknowledgements

The author gratefully acknowledges the many forms of support—intellectual, emotional, and existential—received throughout the long process of developing this work. Some of this support was explicit, while other forms were quiet and unnamed, yet all were deeply meaningful and helped

make this work possible.

References

- [1] L. Xie, “Experiential Noise and Learning: A First-Person Perspective for Noise-Sensitive Learners” EdArXiv, 2026.
- [2] L. Xie, “A First-Person Necessary Condition for Alignment in AI Systems with AGI-like Properties” GitHub repository, 2026.
- [3] L. Xie, “Real-Time Learning as an Experiential Process: A First-Person Perspective on Generality, Creativity, and Alignment” GitHub repository, 2026.

“Nothing is so strong as gentleness.”

— Jane Austen